Cutting Rules for In-In Correlators

& Cosmological Collider Signals

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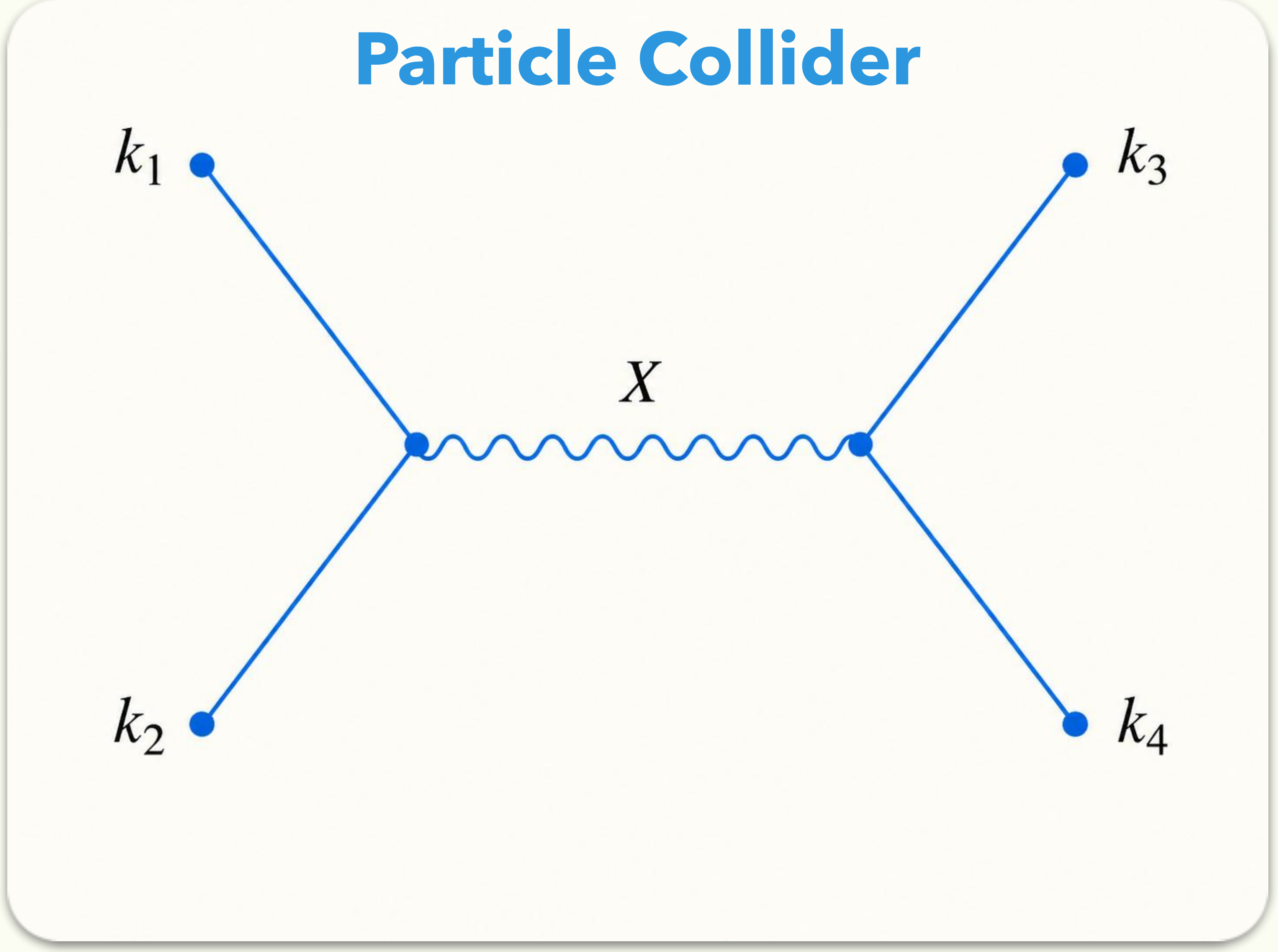
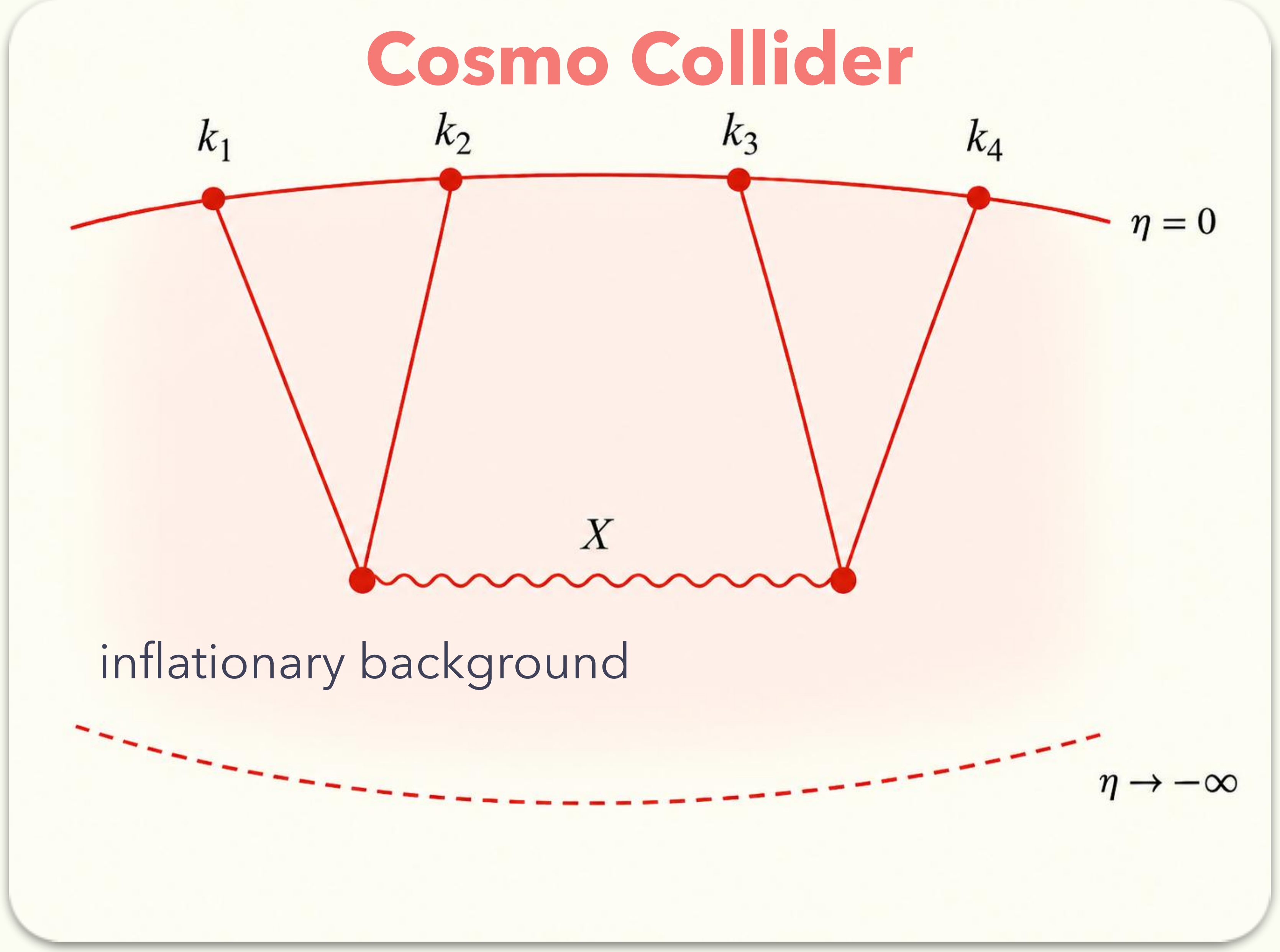
Based on 2409.07521 with Y.Ema

Cosmological Collider?



v.s.



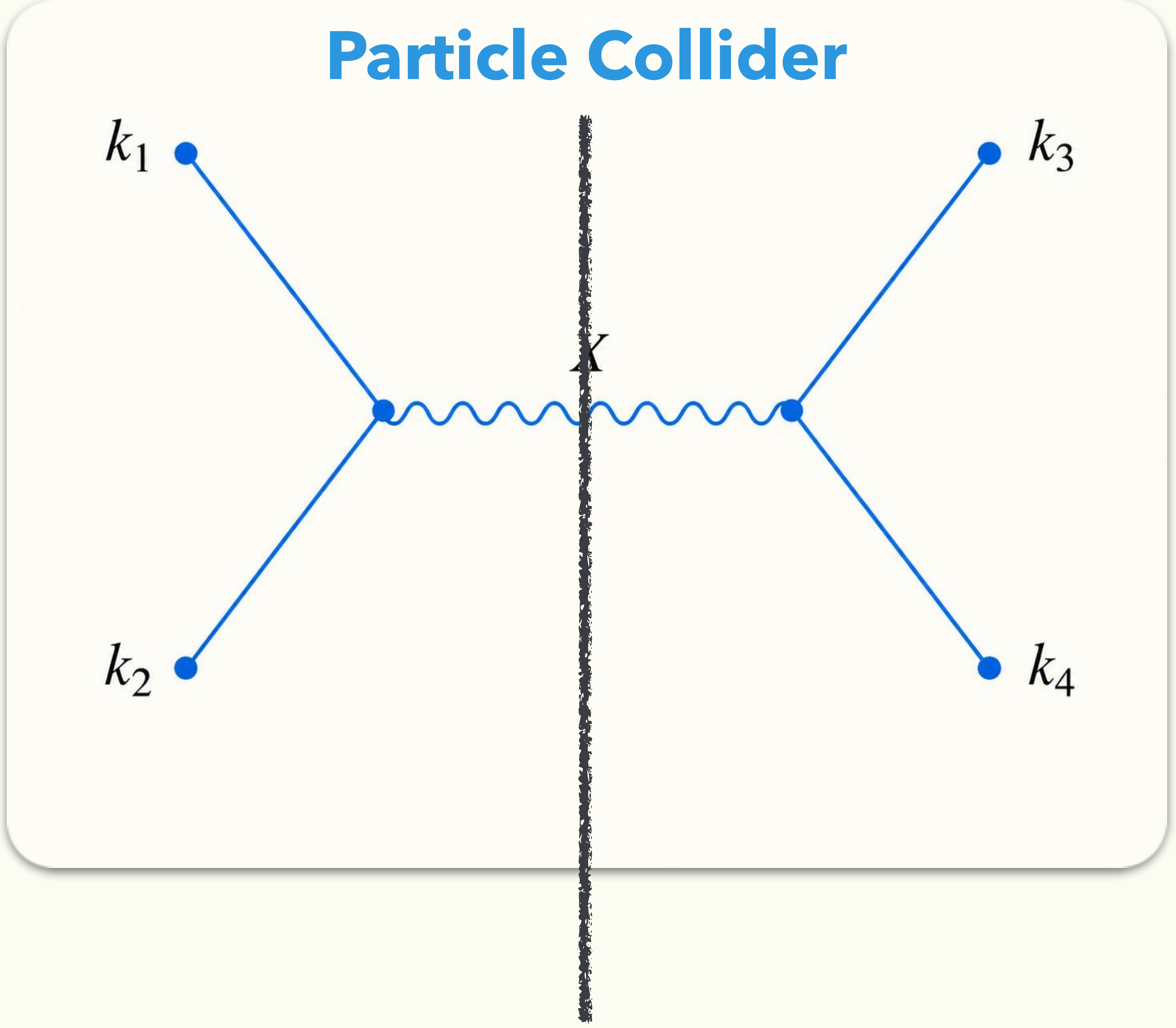
On-shell particle in amplitudes

Vacuum: time-translational inv.

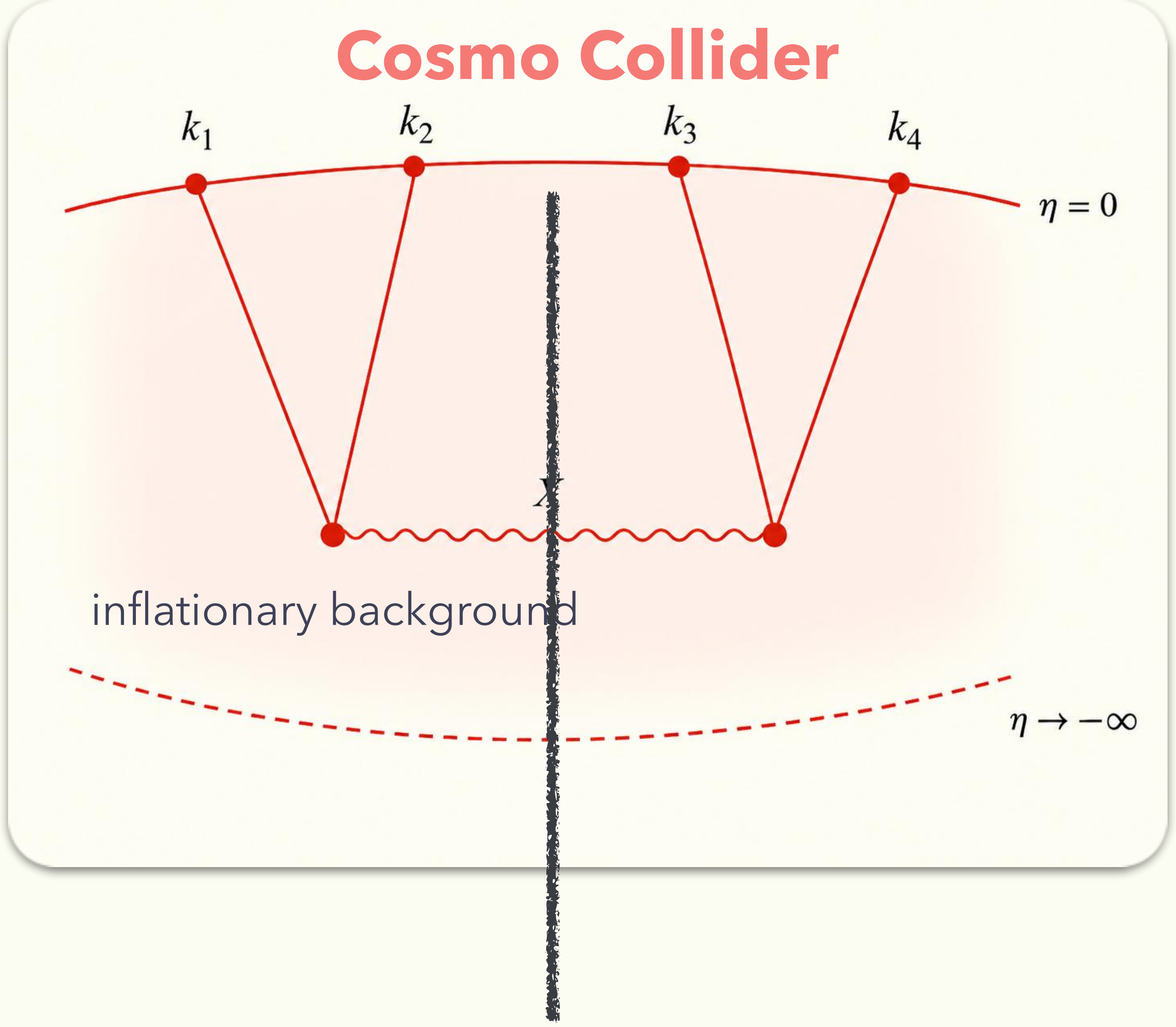
On-shell history in correlators

Inflationary bkg: no time-translational inv.

Cut & Non-analyticity



v.s.



Cut = On-shell intermediate particle

Non-analyticity in amplitudes

?

In-In Diagrams

Keldysh basis & causal flows

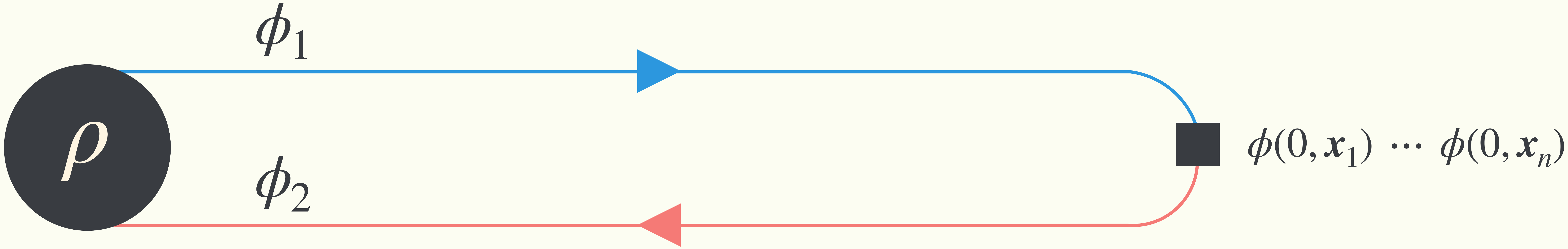
In-In Formalism

- Path-integral on the Keldysh contour

$$\langle \phi(0,\boldsymbol{x}_1)\,\cdots\,\phi(0,\boldsymbol{x}_n)\rangle = \int \mathrm{d}\phi_{1i}\,\mathrm{d}\phi_{2i}\,\mathrm{d}\phi_f\,\langle \phi_{1i},-\infty\,|\,\rho\,|\,\phi_{2i},-\infty\rangle$$

$$\times\,\langle \phi_f,0\,|\,\phi_{1i},-\infty\rangle\langle \phi_{2i},-\infty\,|\,\phi_f,0\rangle \times \phi_f(\boldsymbol{x}_1)\,\cdots\,\phi_f(\boldsymbol{x}_n)$$

$$\int_{\phi_{1i}}^{\phi_f} \mathscr{D}\phi_1\, e^{iS[\phi_1]} \qquad\qquad \int_{\phi_{2i}}^{\phi_f} \mathscr{D}\phi_2\, e^{-iS[\phi_2]}$$



Propagators in In-In Formalism

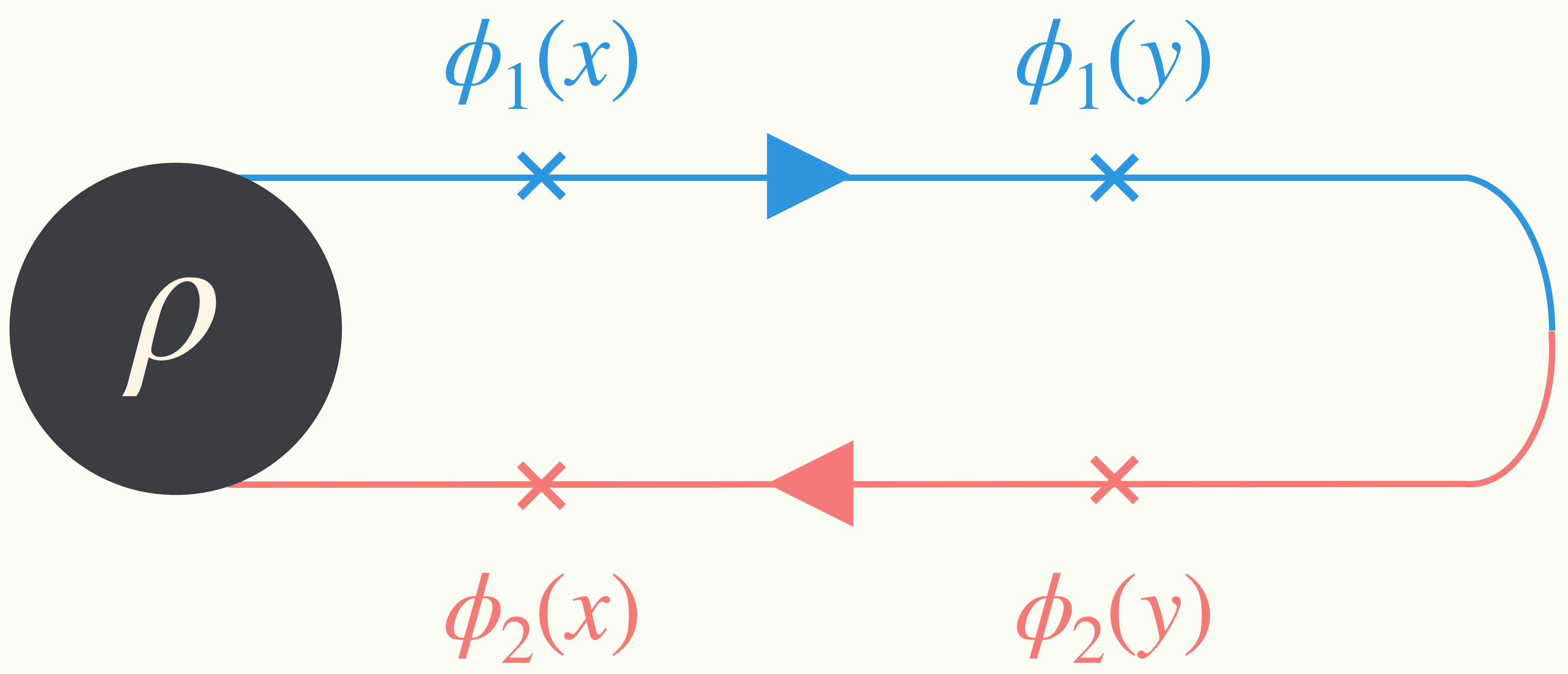
- (Apparently) four propagators in **1/2 basis**

$$G_{11}(x,y) := \langle \mathcal{T}_C\, \phi_1(x)\, \phi_1(y) \rangle = \langle \mathcal{T}\, \phi(x)\, \phi(y) \rangle$$
$$= G_F(x,y) \rightarrow$$ **Feynman propagator**

$$G_{12}(x,y) := \langle \mathcal{T}_C\, \phi_1(x)\, \phi_2(y) \rangle = \langle \phi(y)\, \phi(x) \rangle$$
$$= G_<(x,y) \rightarrow$$ **Wightman negative** func.

$$G_{21}(x,y) := \langle \mathcal{T}_C\, \phi_2(x)\, \phi_1(y) \rangle = \langle \phi(x)\, \phi(y) \rangle$$
$$= G_>(x,y) \rightarrow$$ **Wightman positive** func.

$$G_{22}(x,y) := \langle \mathcal{T}_C\, \phi_2(x)\, \phi_2(y) \rangle = \langle \tilde{\mathcal{T}}\, \phi(x)\, \phi(y) \rangle$$
$$= G_{\bar{F}}(x,y) \rightarrow$$ **Dyson** propagator



Keldysh r/a Basis

- Manifest **causal flow** in **r/a basis**

$$\phi_r \equiv \frac{1}{2}\left(\phi_1 + \phi_2\right), \qquad \phi_a \equiv \phi_1 - \phi_2$$

Statistical function/propagator

$$G_{rr}(x,y) := \langle \phi_r(x)\,\phi_r(y)\rangle = \frac{1}{4}\left(G_F + G_< + G_> + G_{\tilde{F}}\right) = \frac{1}{2}\left[G_>(x,y) + G_<(x,y)\right]$$

Retarded function/propagator

$$G_{ra}(x,y) := \langle \phi_r(x)\,\phi_a(y)\rangle = \frac{1}{2}\left(G_F - G_< + G_> - G_{\tilde{F}}\right) = \theta(x_0, y_0)\left[G_>(x,y) - G_<(x,y)\right]$$

$$G_{aa}(x,y) := \langle \phi_a(x)\,\phi_a(y)\rangle = \frac{1}{2}\left(G_F - G_< - G_> + G_{\tilde{F}}\right) = 0$$

Keldysh r/a Basis

[Keldysh '64]

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Corresponding **causal flow**

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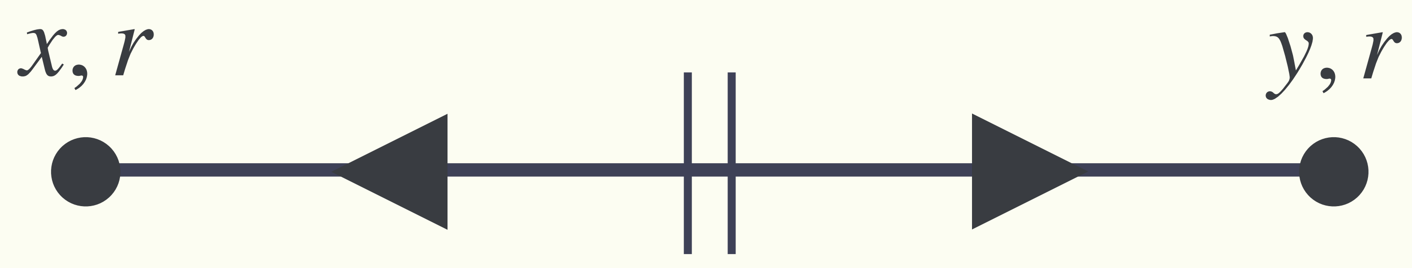

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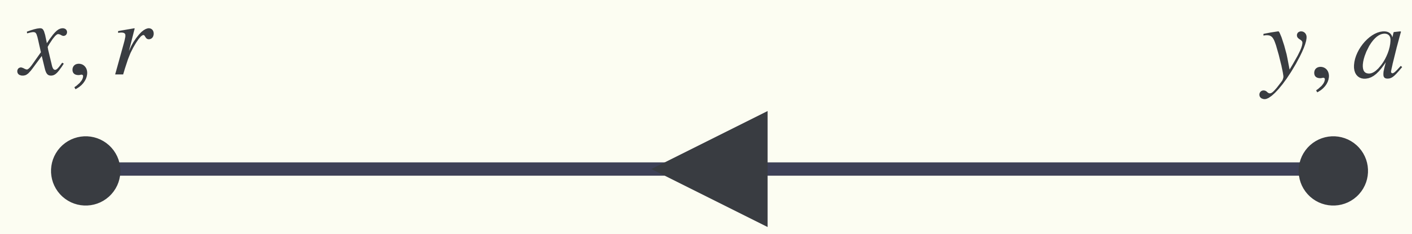

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Keldysh r/a Basis

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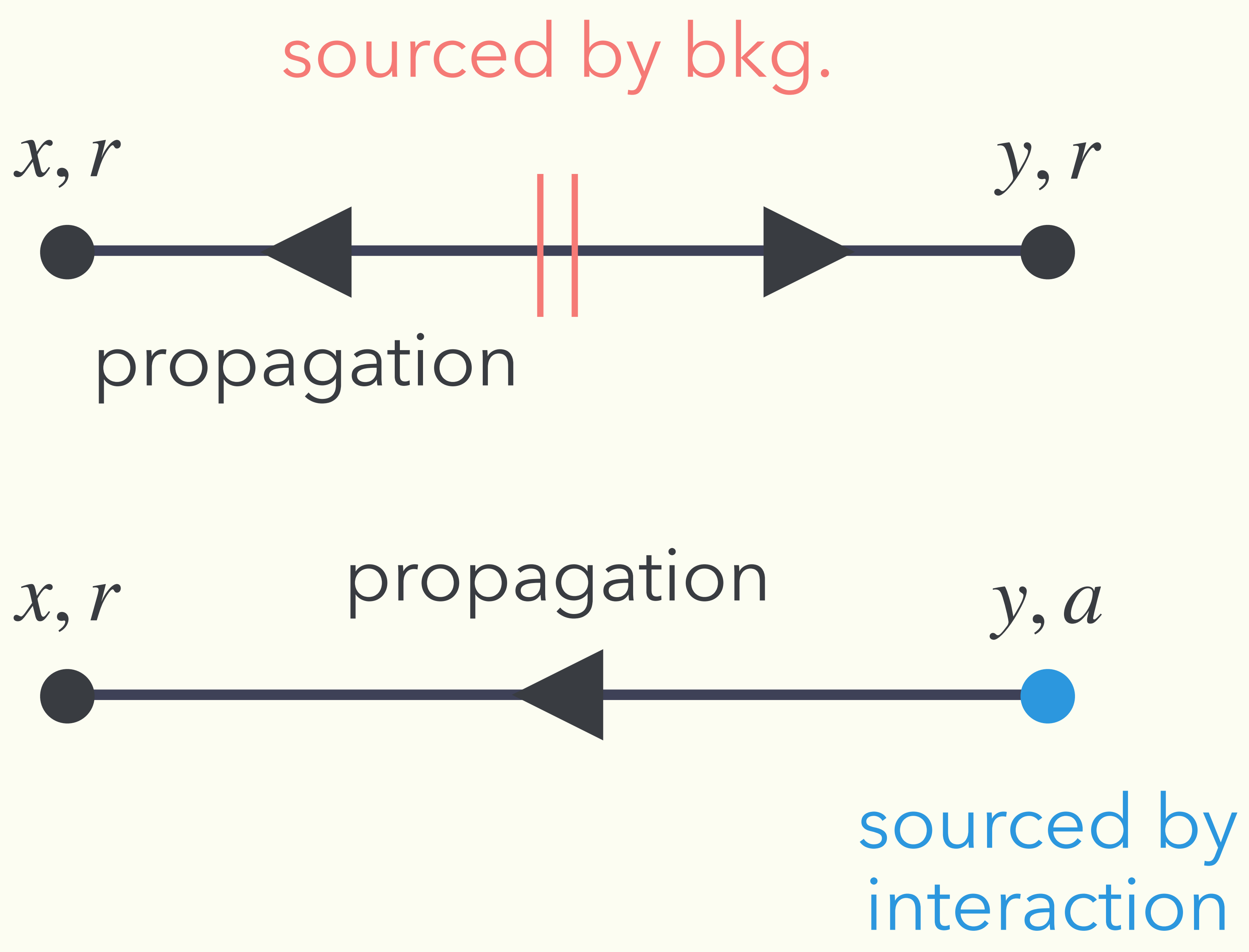
Corresponding **causal flow**

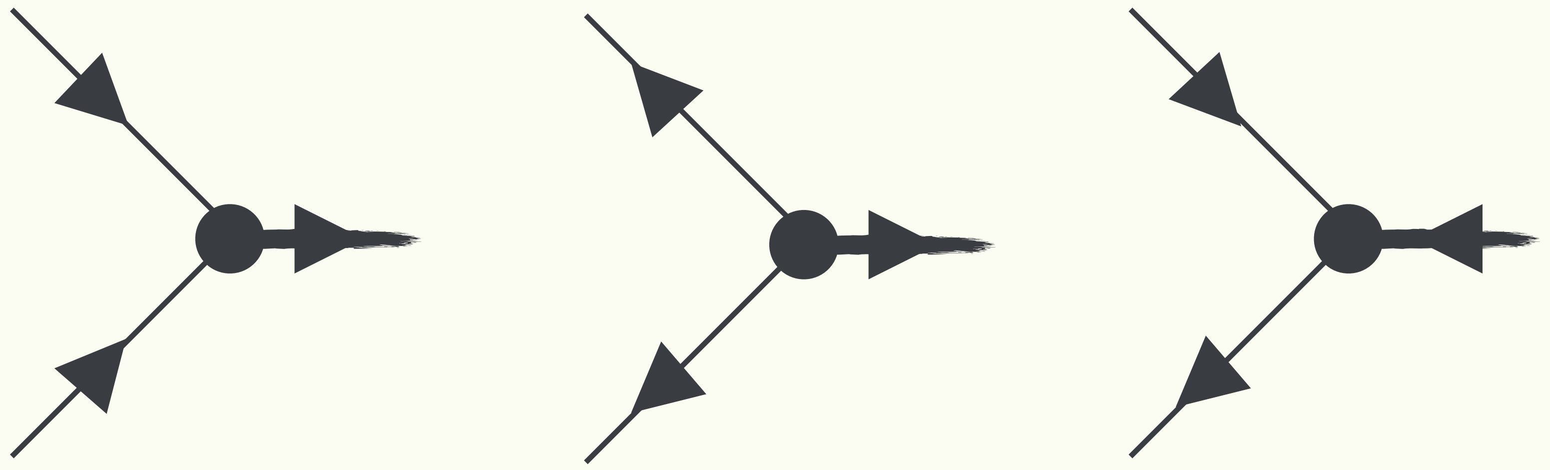


Keldysh r/a Basis

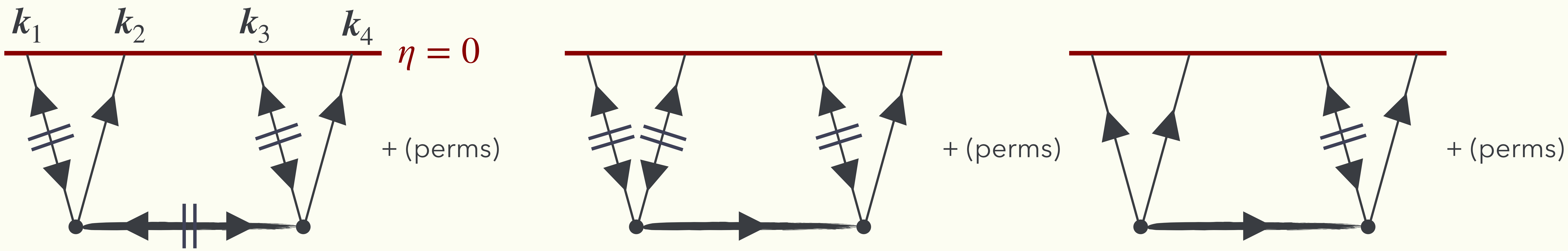
[Keldysh '64]

- Manifest **causal flow** in **r/a basis** (cont'd)
  - E.g., 3pt interaction: $\lambda_3 \left(\partial\phi\right)^2 \sigma$

$$\left(\partial\phi_1\right)^2 \sigma_1 - \left(\partial\phi_2\right)^2 \sigma_2 = \left[\left(\partial\phi_r\right)^2 + \frac{1}{4}\left(\partial\phi_a\right)^2\right] \sigma_a + 2\left(\partial\phi_a\right)\left(\partial\phi_r\right)\sigma_r$$



$$\langle \phi_{\boldsymbol{k}_1}\phi_{\boldsymbol{k}_2}\phi_{\boldsymbol{k}_3}\phi_{\boldsymbol{k}_4}\rangle =$$



But,…many redundant structures. → **Cutting rules** = **causal reorganization**

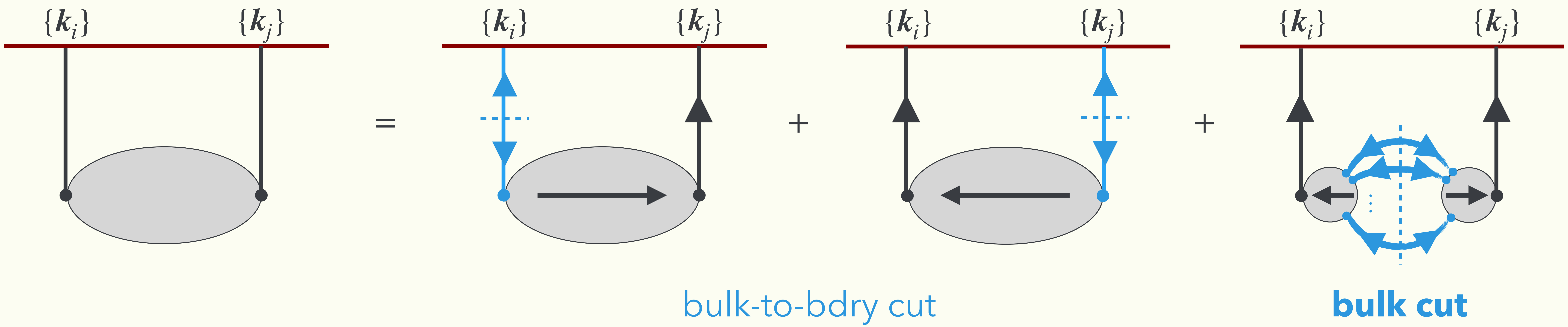
Cutting Rules for In-In

Exact docomposition into causal blocks

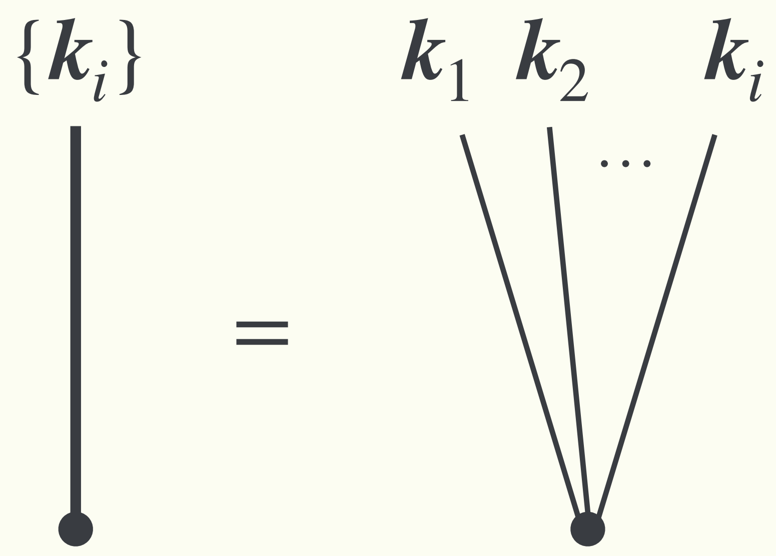
Cutting Rules for In-In Correlators

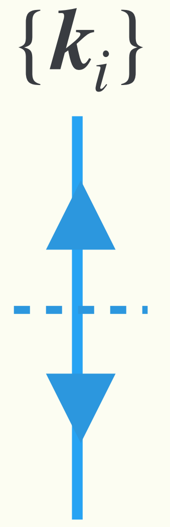
[Caron-Huot '07;
Y. Ema, **KM** 2409.07521]

- Statement for **bulk 2pt function** (npt will be discussed later)



* Notations

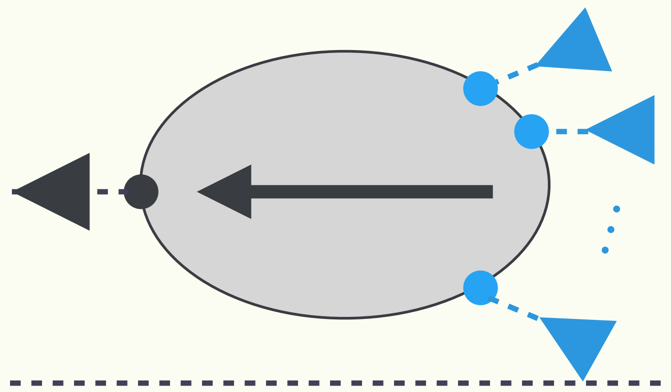




Product of **Wightman funcs.**

$$= \frac{1}{2}\left[\Pi^{i}_{a=1} G_{>}(\tau_1, \tau_2; \boldsymbol{k}_a) + \Pi^{i}_{a=1} G_{<}(\tau_1, \tau_2; \boldsymbol{k}_a)\right]$$

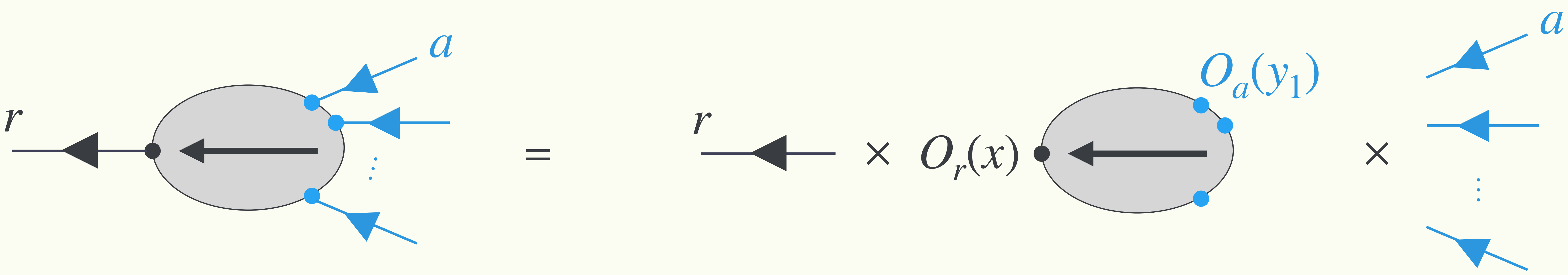
Fully Retarded Func. (FRF)



Fully Retarded Function

[Y. Ema, **KM** 2409.07521]

- Fundamental **causal** building block



$$\mathscr{V}_{ra\cdots a}(x, y_1, \cdots, y_{n-1}) = \langle O_r(x)\, O_a(y_1)\ \cdots\ O_a(y_{n-1})\rangle$$

- **Causal response** of x to sources at $y_1,\ \cdots, y_{n-1}$ (incl. micro-causality)

$\mathscr{V}_{ra\cdots a}(x, y_1, \cdots, y_{n-1}) = 0$ if $^{\exists}y_i$ s.t. $x^0 < y_i^0$ or $\left(x - y_i\right)^2 < 0$ E.g.) $n = 2$ $\mathscr{V}_{ra}(x, y) = \theta(x_0 - y_0)\,\langle[O(x), O(y)]\rangle$

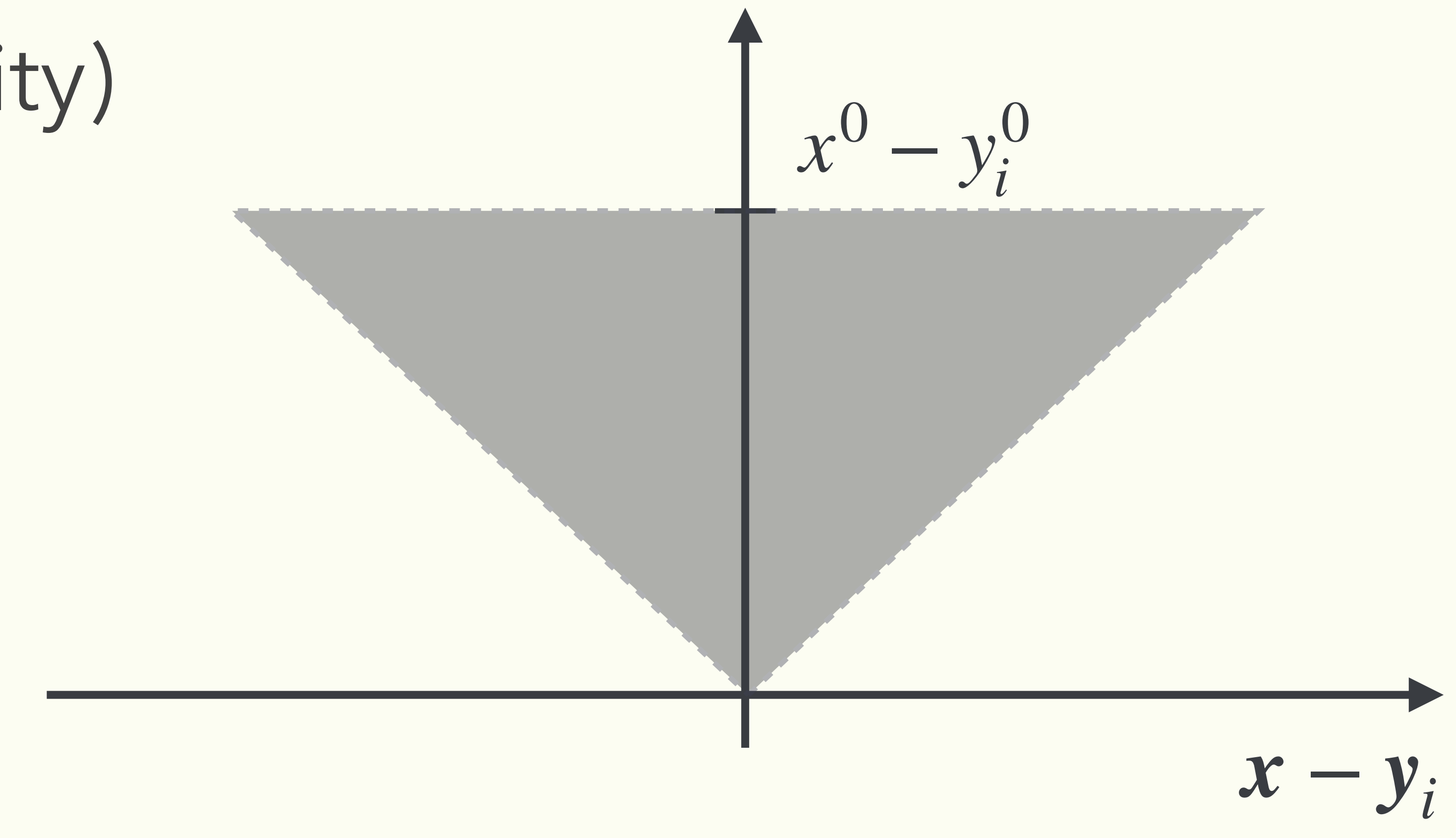
One can show this by mathematical induction starting from $n = 2$. (see for details.)

Fully Retarded Function

[Y. Ema, **KM** 2409.07521]

- Fundamental **causal** building block (cont'd)
  - **Causal response** of x to sources at $y_1, \cdots, y_{n-1}$ (incl. micro-causality)

$$\mathscr{V}_{ra\cdots a}(x, y_1, \cdots, y_{n-1}) = 0 \quad \text{if } {}^{\exists}y_i \text{ s.t. } x^0 < y_i^0 \text{ or } \left(x - y_i\right)^2 < 0$$



→ finite spatial support of $\boldsymbol{x} - \boldsymbol{y}_i$ for a given x^0, y_i^0

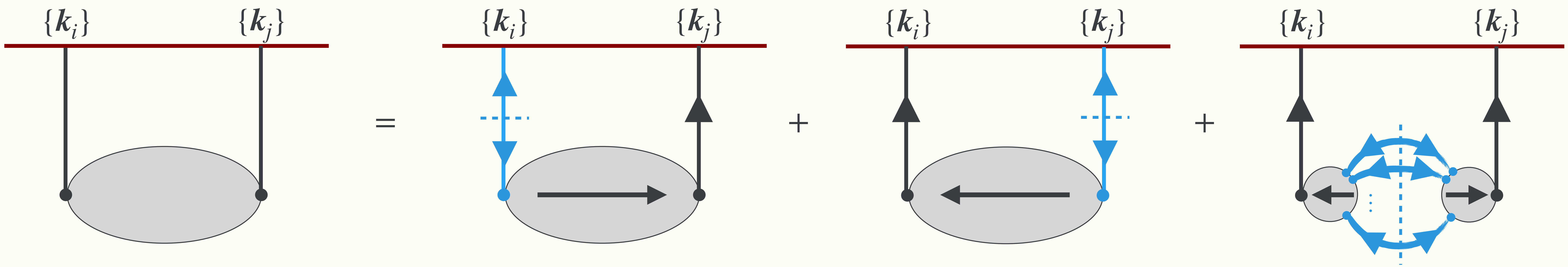
$$\xrightarrow{\text{Fourier tr. wrt } \boldsymbol{y}_i - \boldsymbol{x}} \mathscr{V}_{ra\cdots a}(x^0, y_1^0, \cdots, y_{n-1}^0; \boldsymbol{k}_1, \cdots, \boldsymbol{k}_{n-1})$$

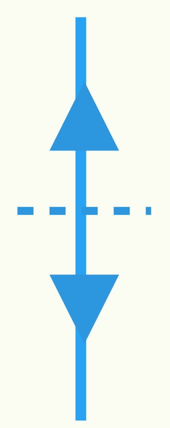
is **analytic** w.r.t. spatial momentum $\boldsymbol{k}_1, \cdots, \boldsymbol{k}_{n-1}$

Cutting Rules for In-In Correlators

[Y. Ema, **KM** 2409.07521]

- Statement for **bulk 2pt function** (see our paper for how to cope with npt)



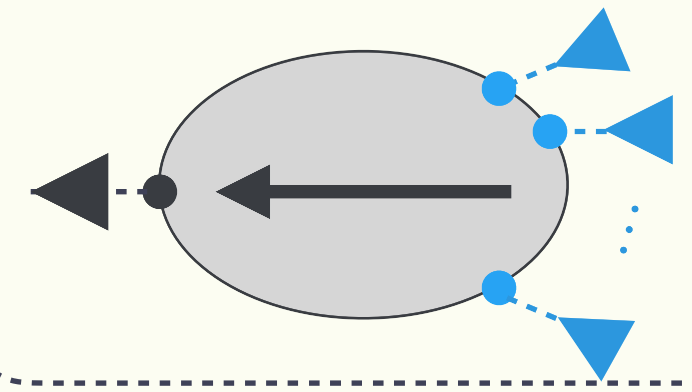


Product of **Wightman funcs.**

incl. **non-analyticity** w.r.t. $\boldsymbol{k}_\bullet$

→ Clear sign of **particle production**!

Fully Retarded Func. (FRF)



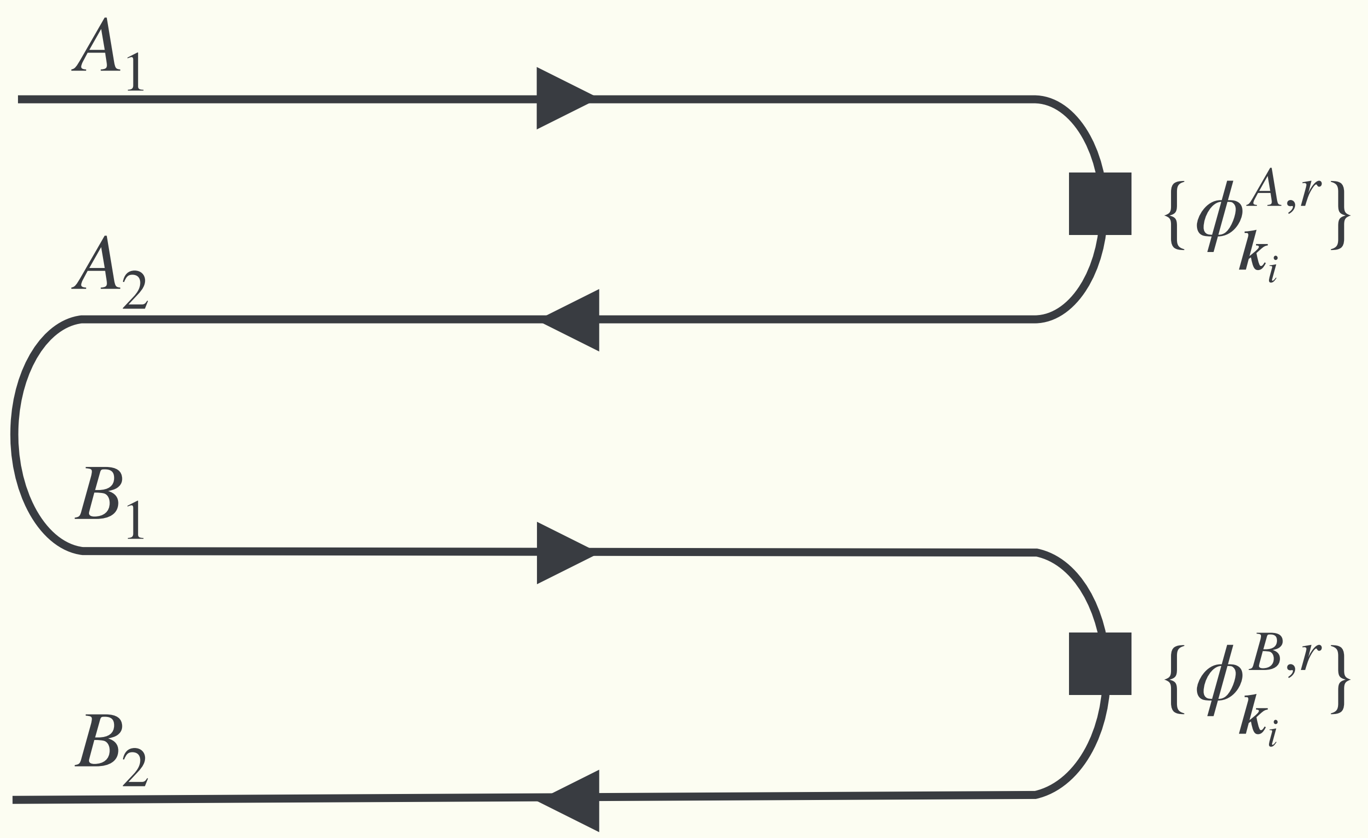
Analytic w.r.t. $\boldsymbol{k}_\bullet$

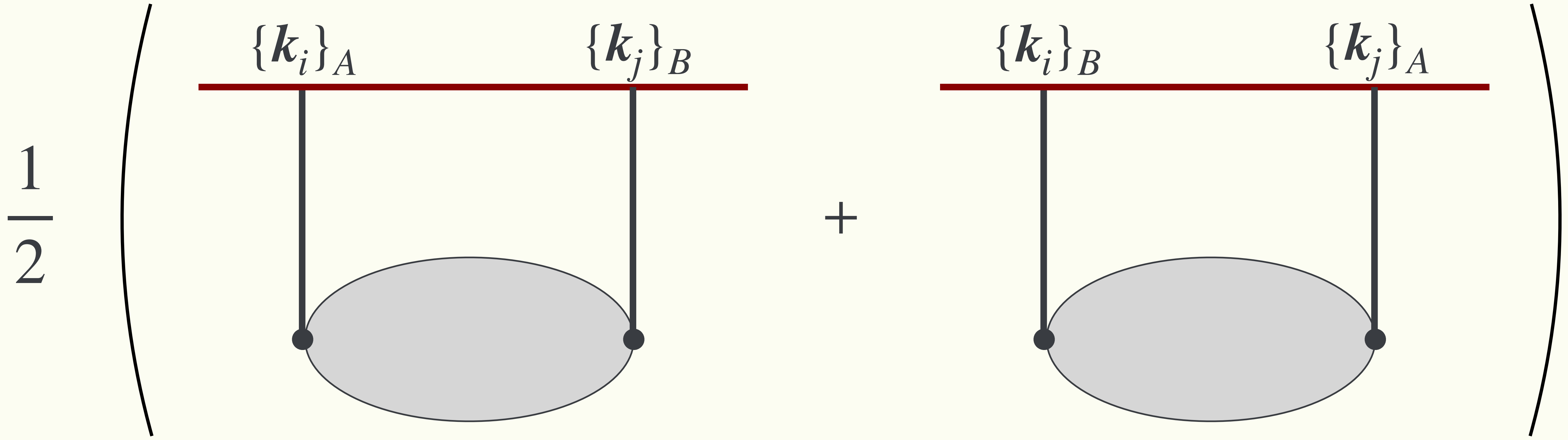
→ causal propagation

Cutting Rules for In-In Correlators

[Caron-Huot '07;
Y. Ema, **KM** 2409.07521]

- Sketch of proof for cutting rule w/ **bulk 2pt function**
  - 1. Add a redundant contour, which classifies inserted **bdry operators**.



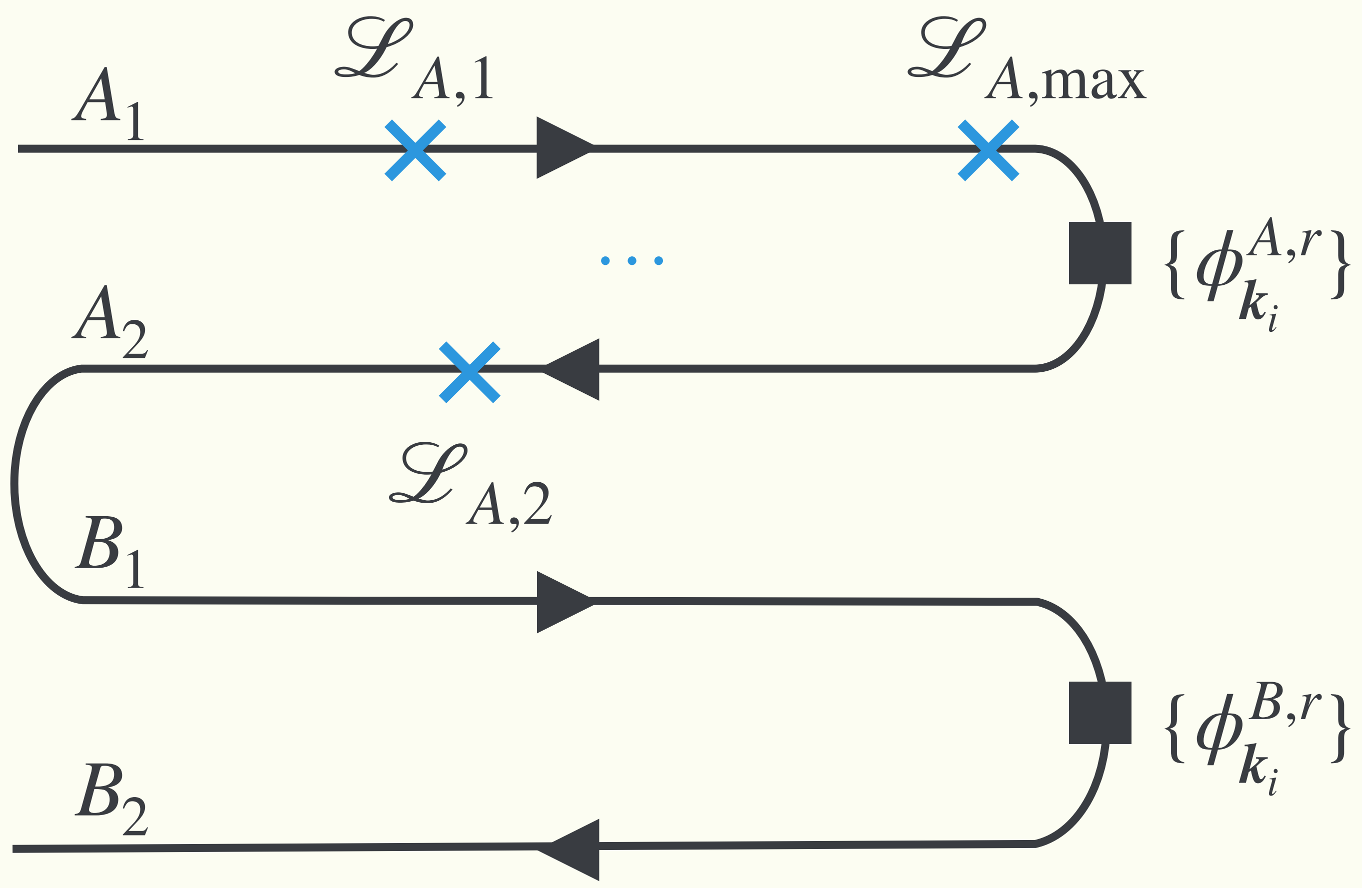


*) Regarded as an insertion of unity $\because \;\; U^\dagger_{A_2} U_{B_1} = \mathbf{1}$

Cutting Rules for In-In Correlators

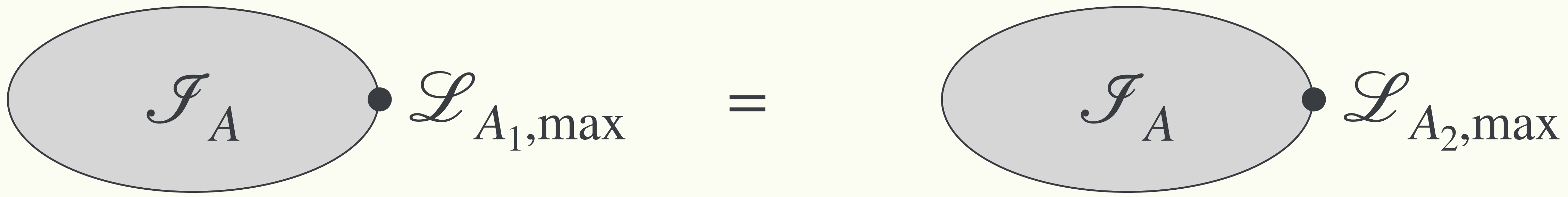
[Caron-Huot '07; Y. Ema, **KM** 2409.07521]

- Sketch of proof for cutting rule w/ **bulk 2pt function**
  - 2. Interaction vertices classified into A/B → **No isolated islands** of A/B-interactions

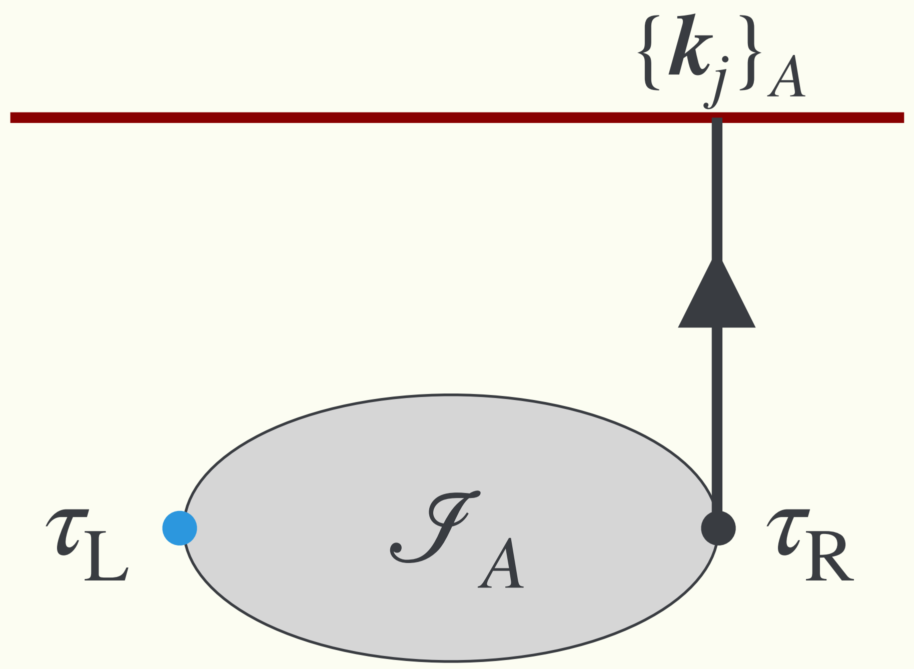


Island: $\mathscr{I}_A \equiv \{\mathscr{L}_{A,i}\,|\,\text{connected}\}$

$\mathscr{L}_{A,\max}$ ($\equiv$ largest time in $\mathscr{I}_A$) is independent of A_1 or A_2



$\mathscr{L}_{A,\max}$ must be connected to the **bdry A-operator**
otherwise $\mathscr{I}_A$ with $\mathscr{L}_{A_1,\max}$ and $\mathscr{L}_{A_2,\max}$ cancel out.

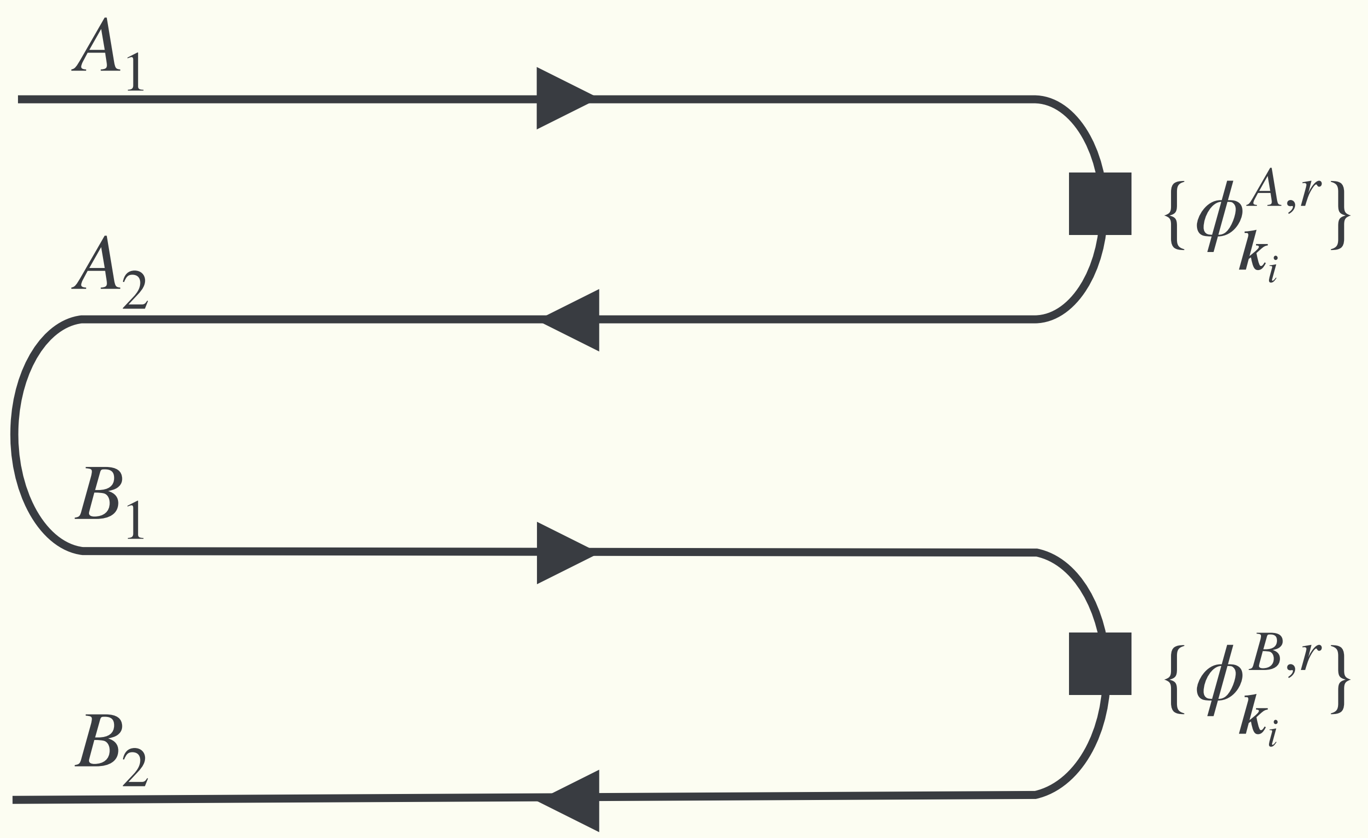


w/ $\tau_R > \tau_L$

Cutting Rules for In-In Correlators

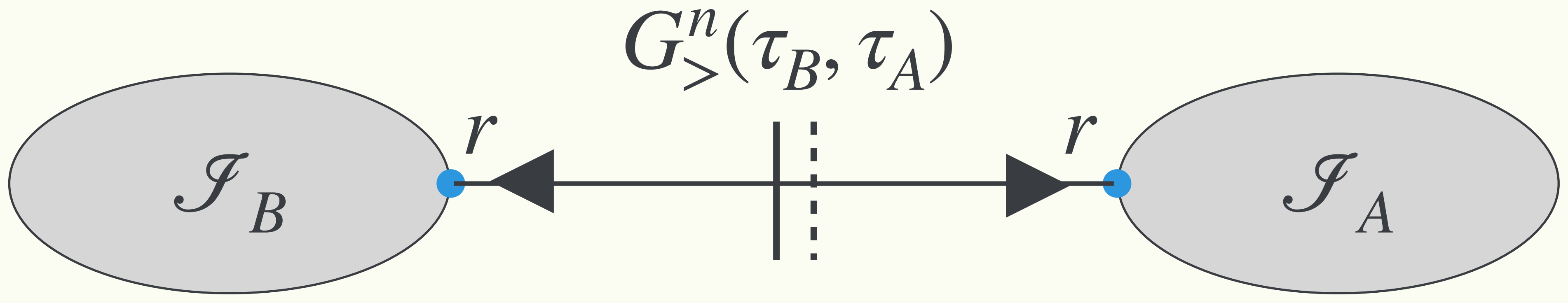
[Caron-Huot '07;
Y. Ema, **KM** 2409.07521]

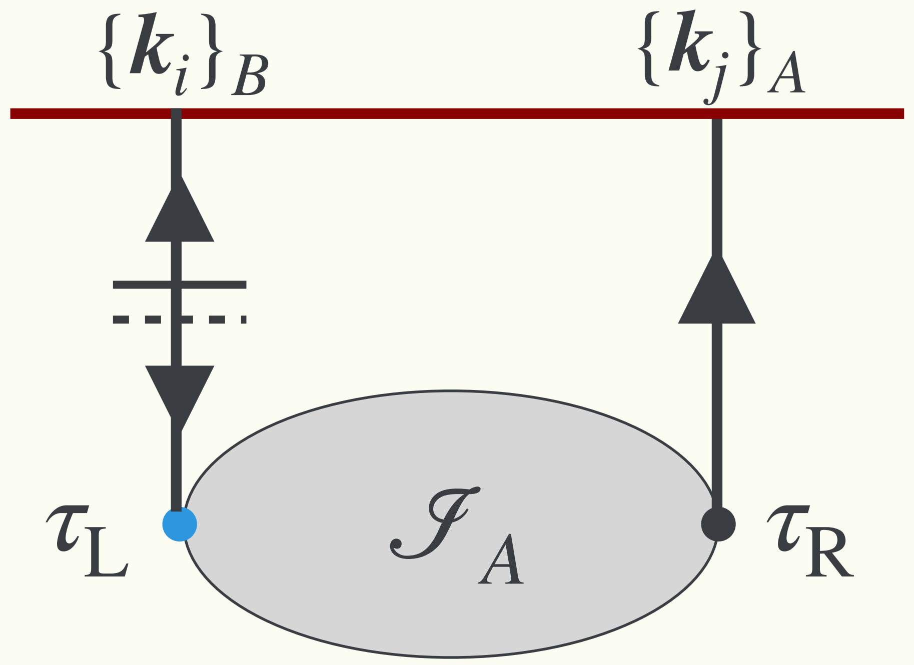
- Sketch of proof for cutting rule w/ **bulk 2pt function**
  - 3. **Cut** appears as an interface of $\mathscr{I}_A$ and $\mathscr{I}_B$



Interface is always connected by $G_{>/<}$ regardless of 1/2.

→ Both connecting points must be r-type operators, $O_{A,r}$ or $O_{B,r}$





w/ $\tau_R > \tau_L$

Cut diagram = product of **Wightman funcs**.

Cutting Rules for In-In Correlators

[Caron-Huot '07;
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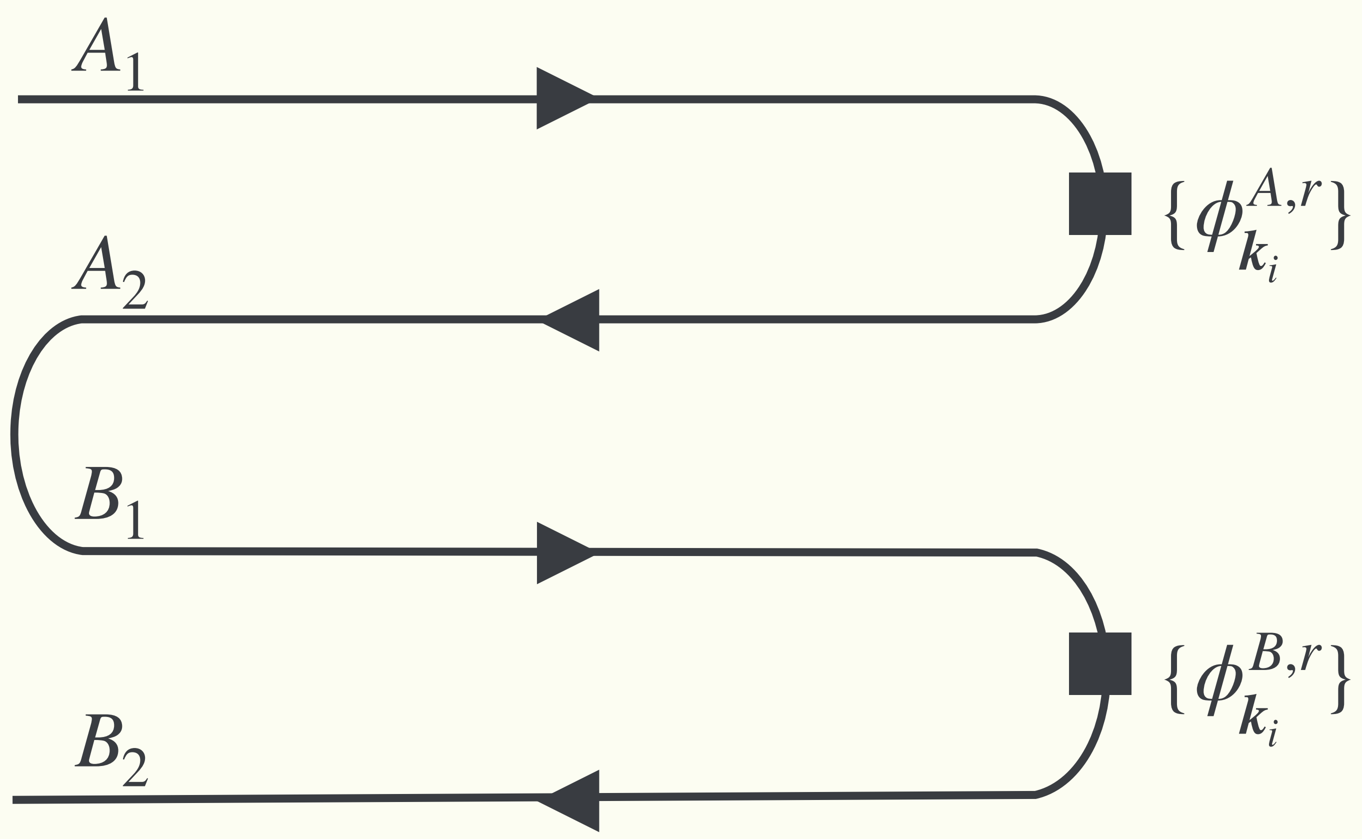

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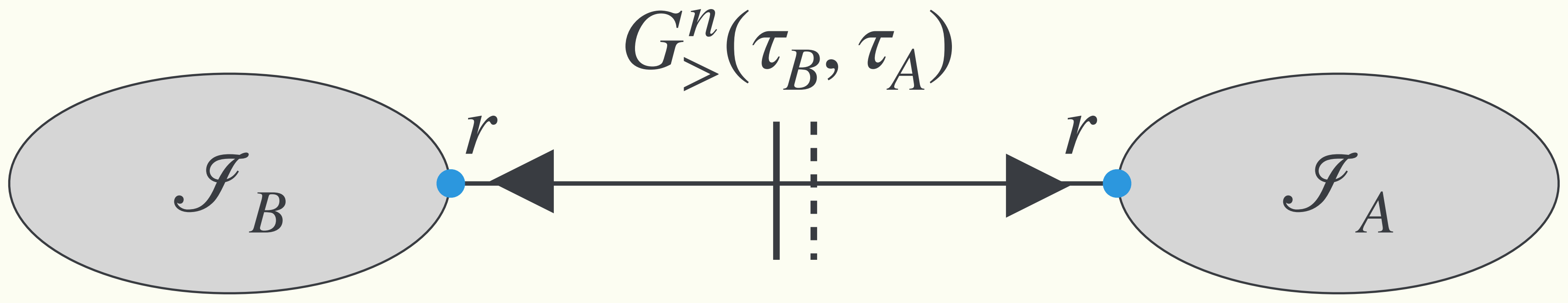
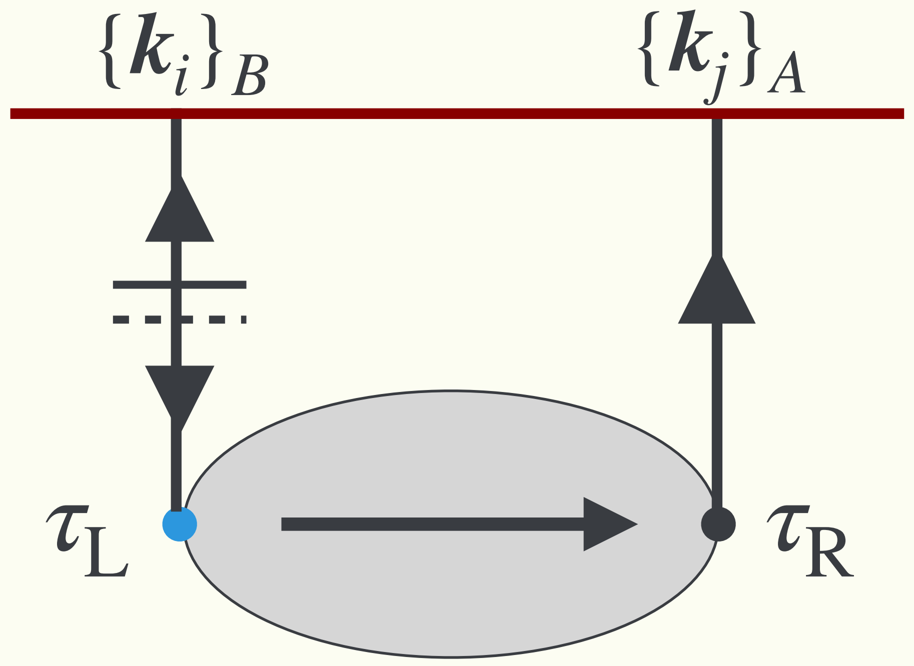




w/ $\tau_R > \tau_L$

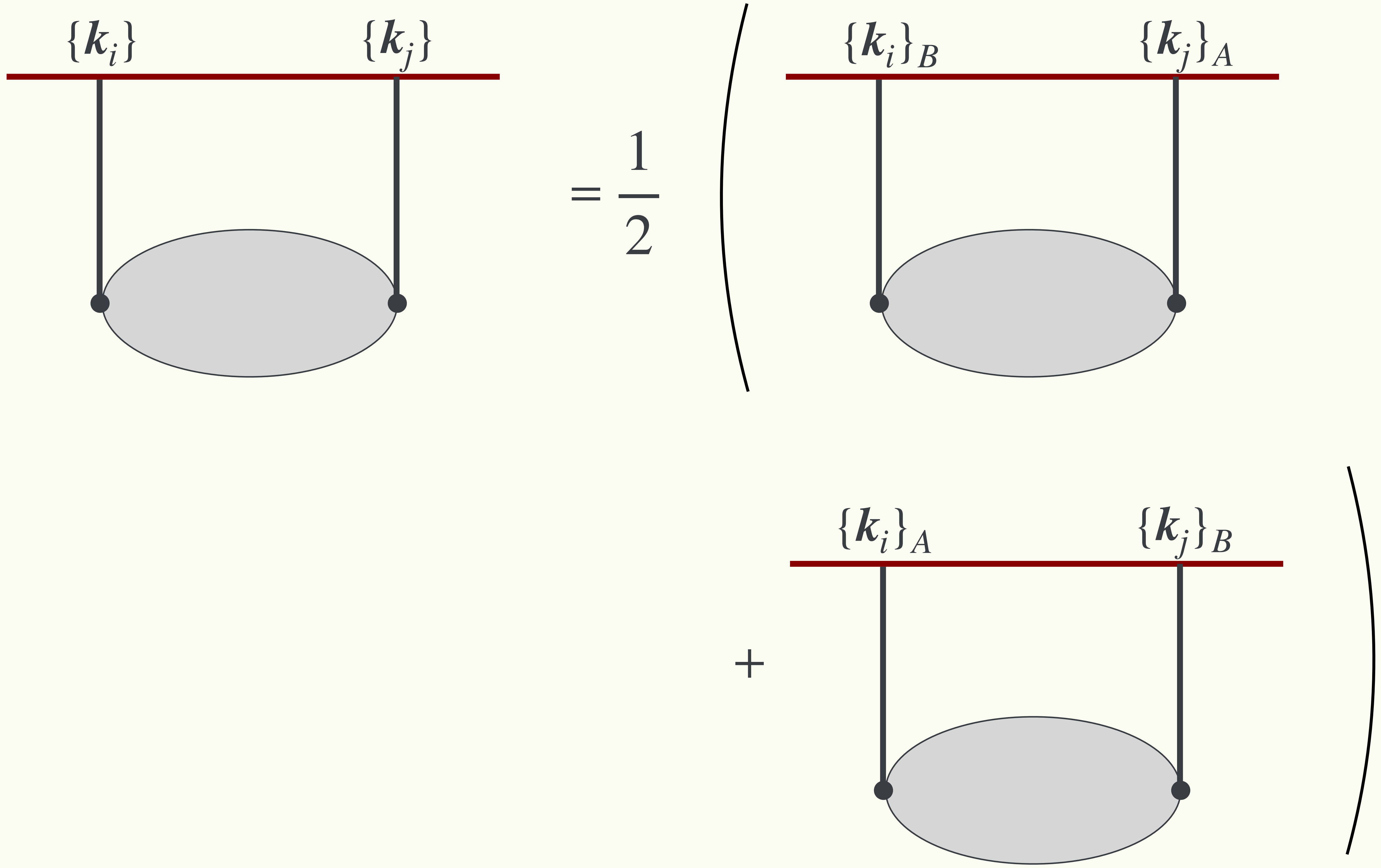
Cut diagram = product of **Wightman funcs**.

Response is encoded in **fully retarded func**.

Cutting Rules for In-In Correlators

[Caron-Huot '07;
Y. Ema, **KM** 2409.07521]

- Sketch of proof for cutting rule w/ **bulk 2pt function**



Cutting Rules for In-In Correlators

[Caron-Huot '07; Y. Ema, **KM** 2409.07521]

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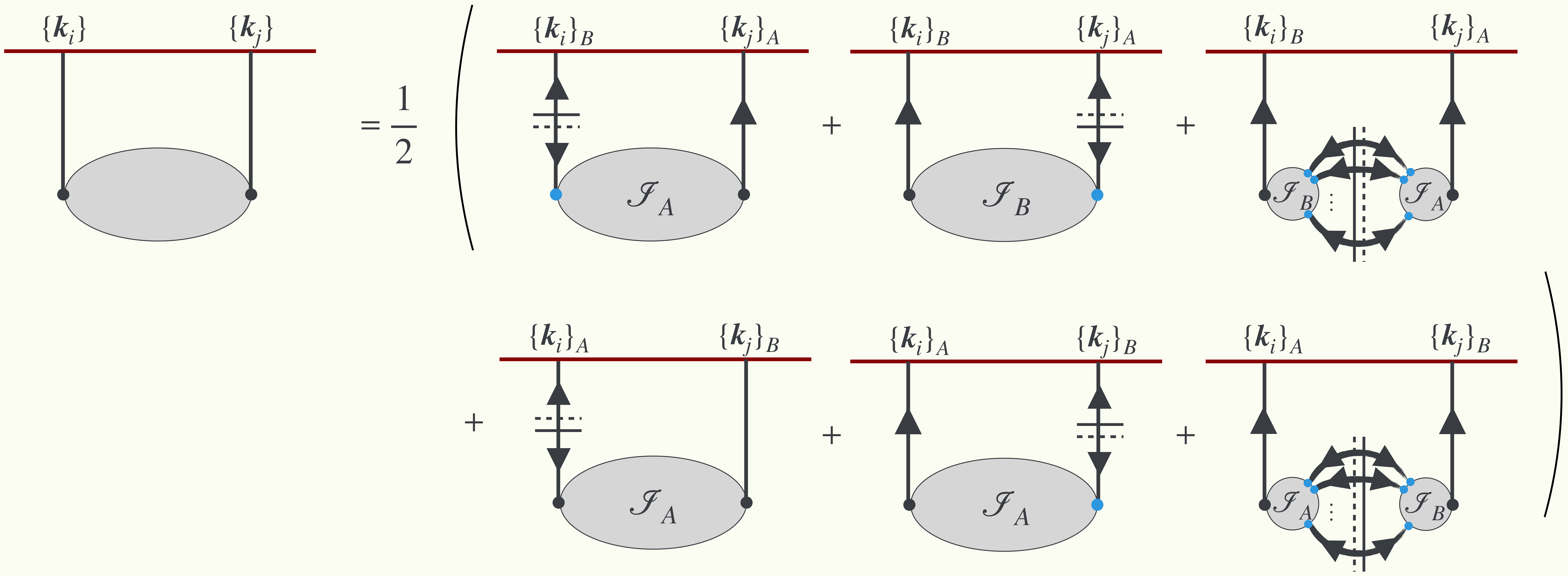

Cutting Rules for In-In Correlators

[Caron-Huot '07;
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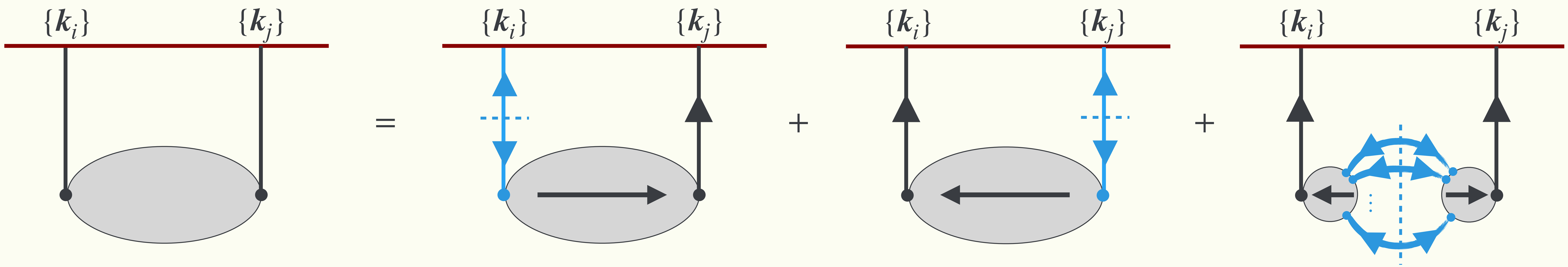
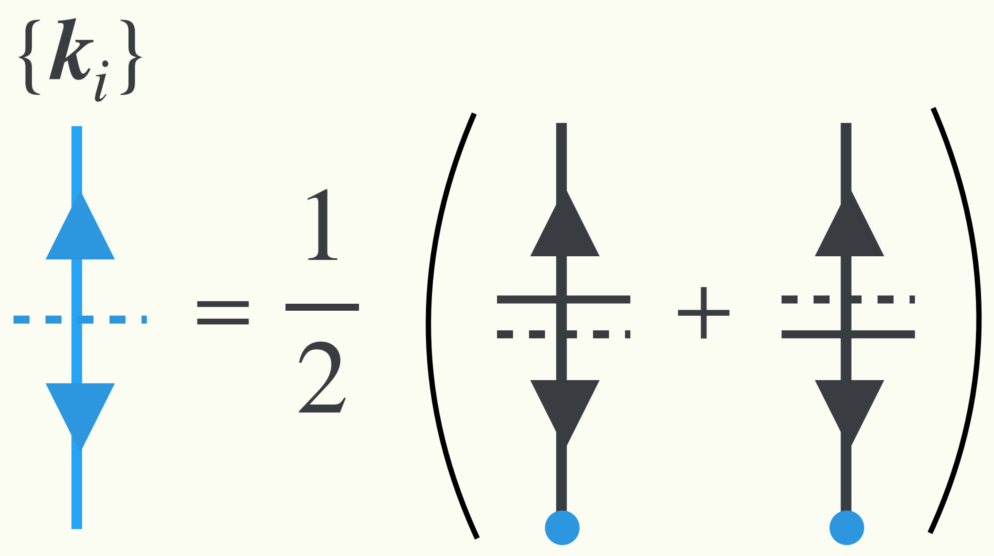


bulk-to-bdry cut

bulk cut

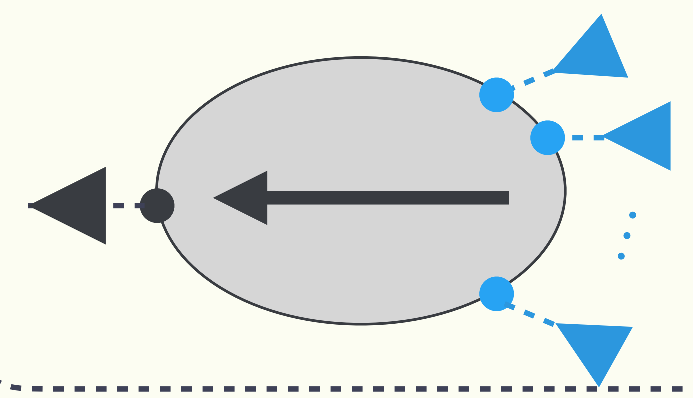


Product of **Wightman funcs.**

incl. **non-analyticity** w.r.t. $\boldsymbol{k}$.

→ Clear sign of **particle production**!

Fully Retarded Func. (FRF)



Analytic w.r.t. $\boldsymbol{k}$.

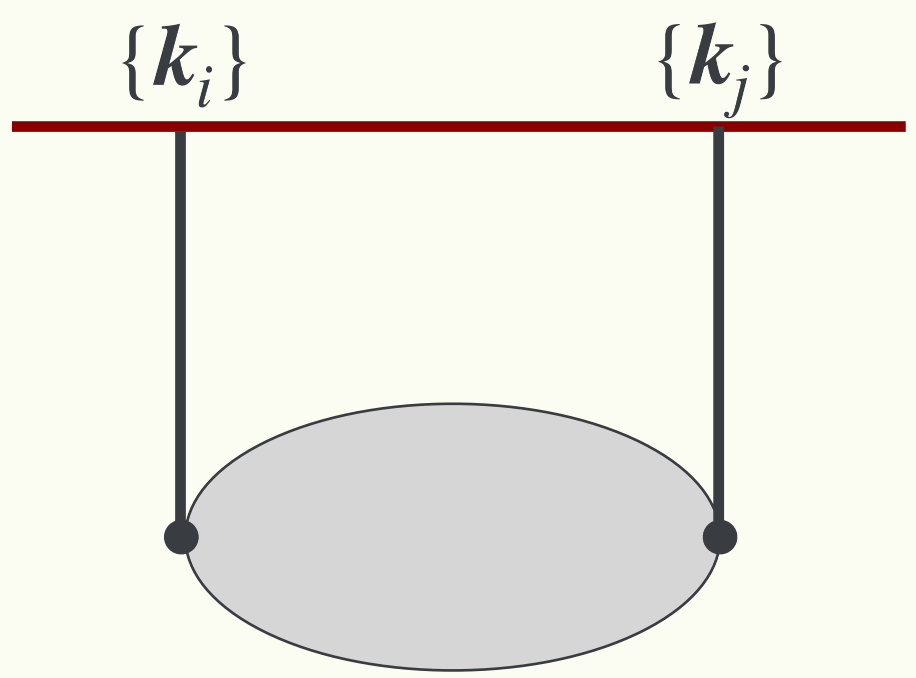
→ causal propagation

Cosmo-Collider Signal

Bulk cut and non-local CC signal

Two types of CC Signals

- Spatially non-local v.s. local CC signals



$$\propto \left(\frac{k_s}{k_{L/R}}\right)^{\pm\nu}, \quad \left(\frac{k_L}{k_R}\right)^{\pm\nu}$$

Non-local Local

Energy sum Vector sum

$$k_L = \sum k_i\,,\ k_R = \sum k_j\,; \quad k_s = \left|\sum \boldsymbol{k}_i\right|$$

$$\nu \equiv \sqrt{\frac{9}{4} - \frac{m^2}{H^2}}$$ → **non-analytic** in momentum

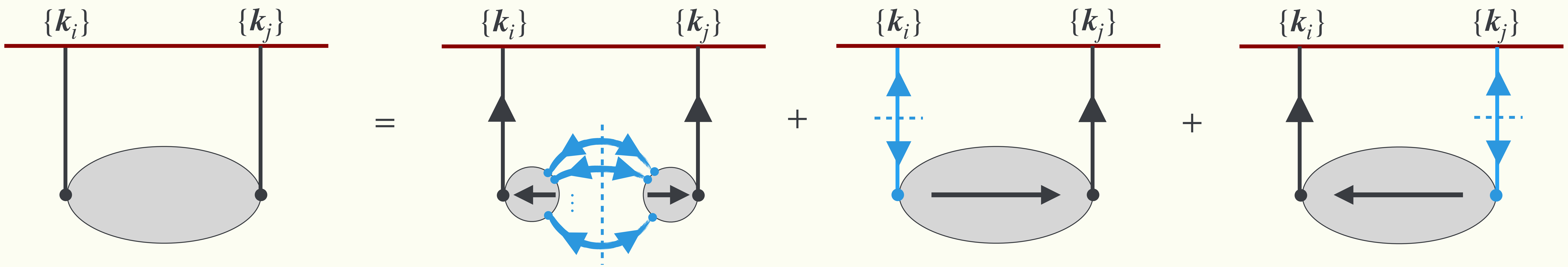
- c.f.) Higher-dimensional operators are always analytic in momentum.

$\mathscr{L} \supset (\partial\phi)^{2n} \longrightarrow k^{2n}$ always integer power of momentum, i.e., analytic.

Non-local CC Theorem

[Y. Ema, **KM** 2409.07521]

- Non-local CC signal solely stems from **bulk cut diagrams**

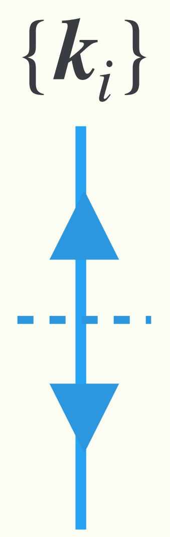


bulk cut

$\supset$ Non-analytic in $\boldsymbol{k}_s$

bulk-to-bdry cut

Analytic in $\boldsymbol{k}_s$ $\because$ FRF

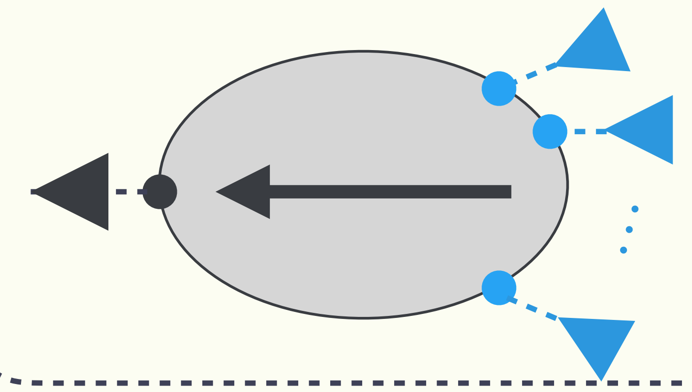


Product of **Wightman funcs.**

incl. **non-analyticity** w.r.t. $\boldsymbol{k}_s$

→ Clear sign of **particle production**!

Fully Retarded Func. (FRF)



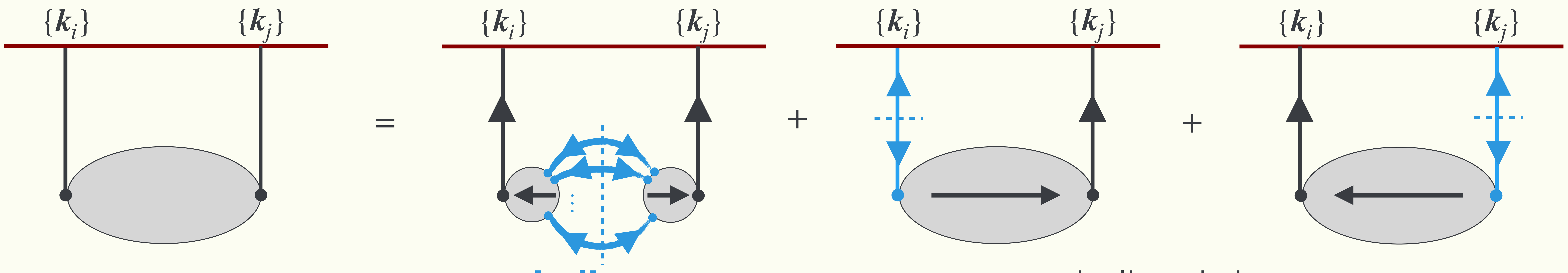
Analytic w.r.t. $\boldsymbol{k}_s$

→ causal propagation

Non-local CC Theorem

[Y. Ema, **KM** 2409.07521]

- Non-local CC signal solely stems from **bulk cut diagrams**



bulk cut
$\supset$ Non-analytic in $\boldsymbol{k}_s$

bulk-to-bdry cut
Analytic in $\boldsymbol{k}_s$ $\because$ FRF

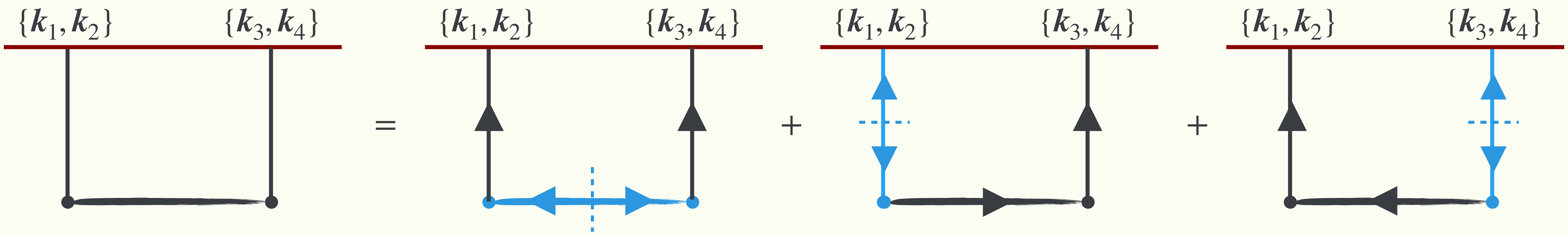
- Our derivation solely depends on basic properties; such as unitarity, locality, and in-in structure.
- It holds theories with **any** fields, interactions, background evolution at **any loop order**.

See e.g., [Goodhew+ '20; Jazayeri+ '21; Xianyu+ '23; Pajer '24]

Non-local CC Theorem

[Y. Ema, **KM** 2409.07521]

- Non-local CC signal solely stems from **bulk cut diagrams**
  - E.g., 3pt interaction: $\lambda_3\, \phi'^2\, \sigma$



$G_{rr}(\tau_L, \tau_R; k_s)$ is non-analytic wrt k_s
→ **non-local** CC

$G_{ra}(\tau_L, \tau_R; k_s)$ is analytic wrt k_s
→ local CC

$$G_{rr}(\tau_L, \tau_R; k_s) \propto \left[J_\nu(-k_s\tau_L)\, J_\nu(-k_s\tau_R) - \cos(\pi\nu) J_\nu(-k_s\tau_L)\, J_{-\nu}(-k_s\tau_R) + (\nu \to -\nu)\right]$$

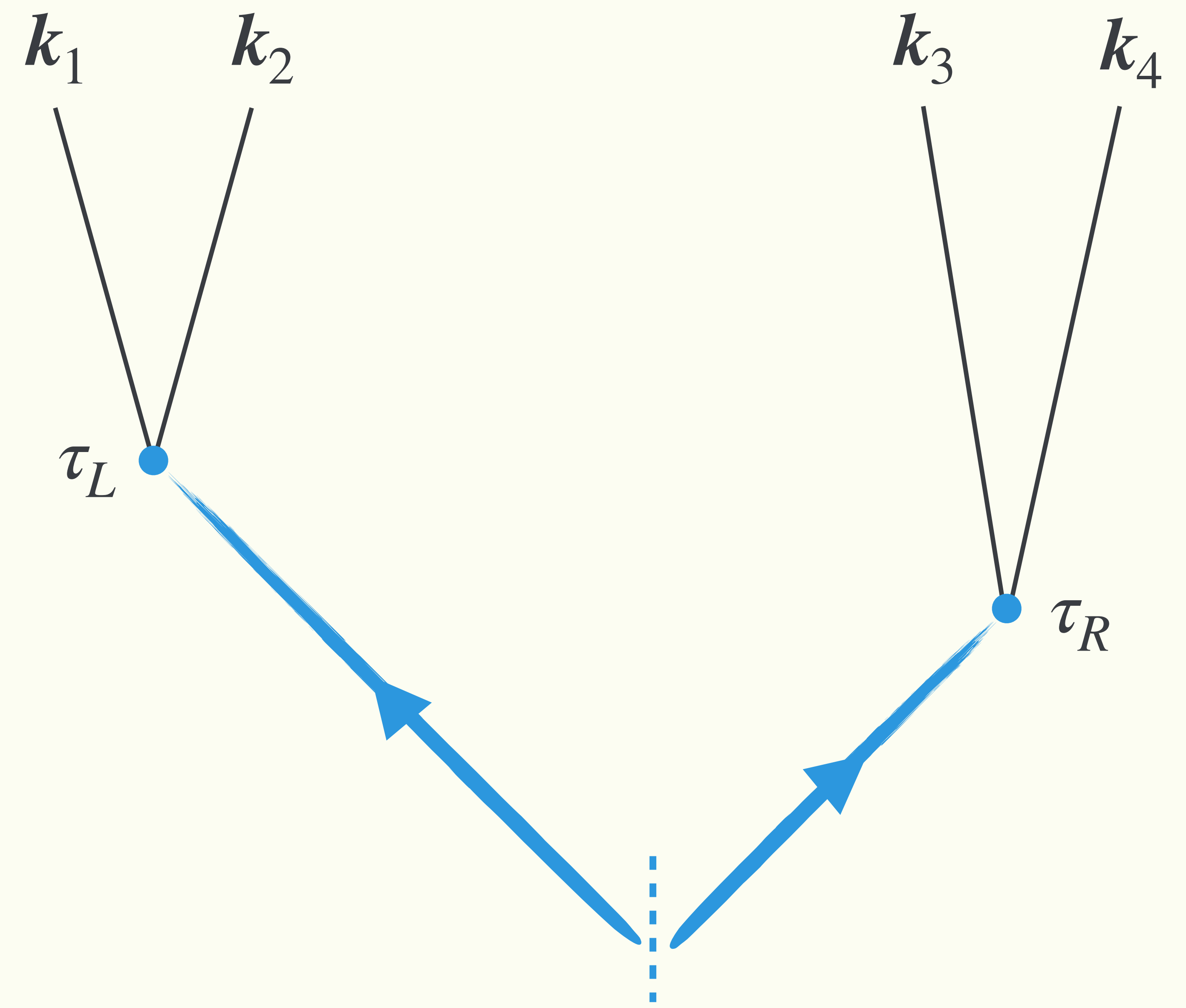
$$G_{ra}(\tau_L, \tau_R; k_s) \propto \left[J_\nu(-k_s\tau_L)\, J_{-\nu}(-k_s\tau_R) - J_{-\nu}(-k_s\tau_L)\, J_\nu(-k_s\tau_R)\right]$$

w/ $J_{\pm\nu}(-k_s\tau) \xrightarrow{k_s \to e^{2\pi i}k_s} e^{\pm 2\pi i\nu}\, J_{\pm\nu}(-k_s\tau)$

Non-local CC Theorem (Intuitive understanding)

- Non-local CC signal solely stems from **bulk cut diagrams**

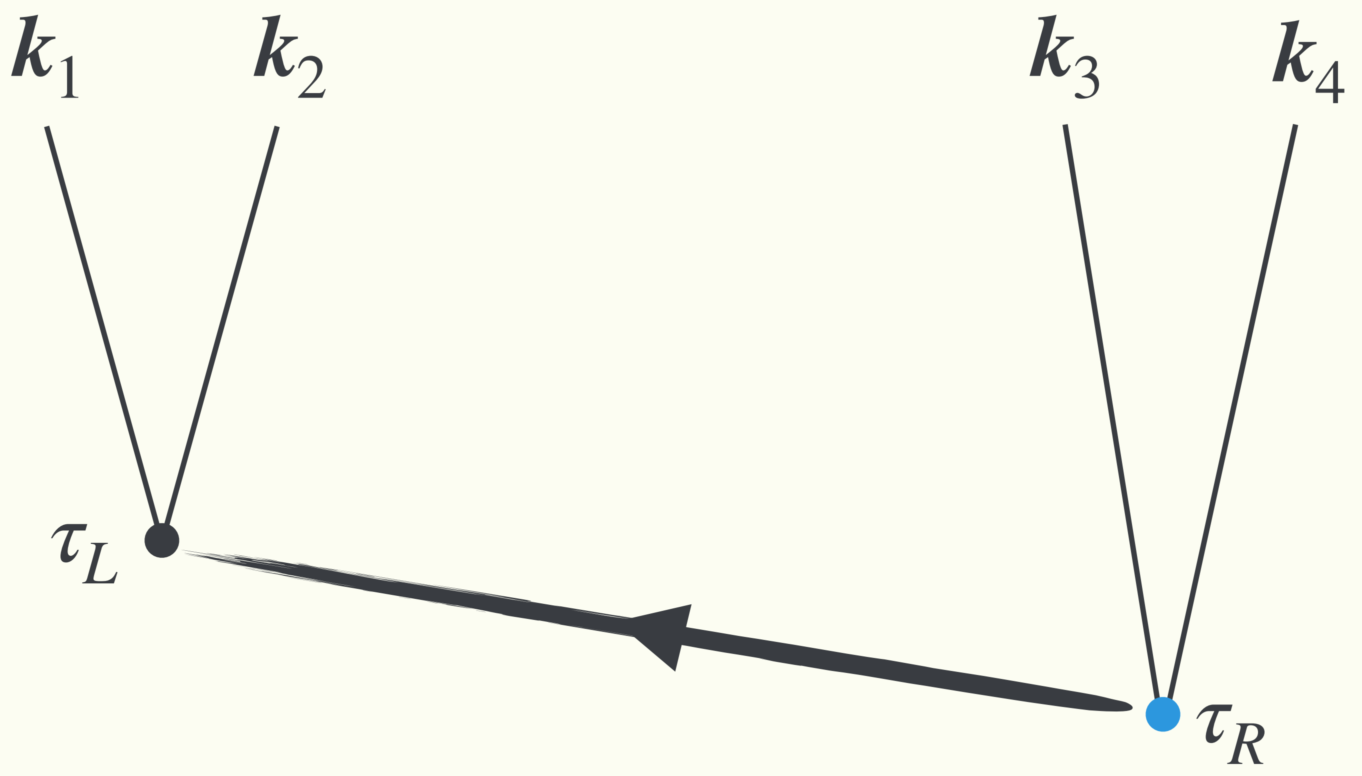
Pair-production from bkg → non-local



Pair-production from bkg

Accumulation → **non-local**

Propagation from localized source → local



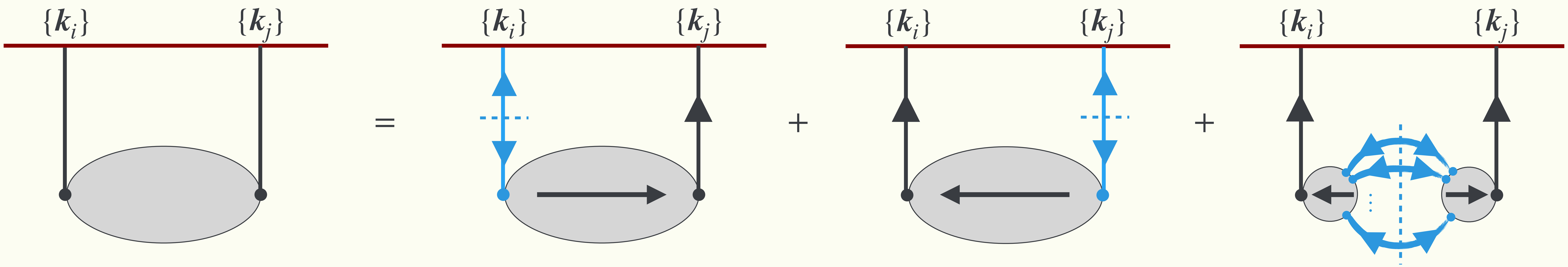
Causal propagation from τ_R to τ_L

Source is localized → local

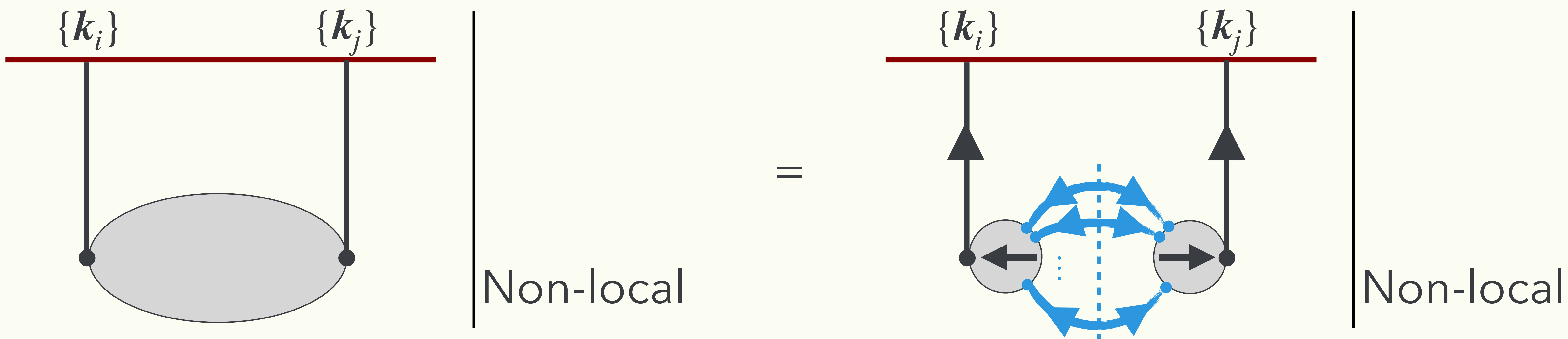
Summary

[Y. Ema, **KM** 2409.07521]

- General cutting rule for in-in = **Cut** (Wightman funcs.) × **FRF** (analytic in $\boldsymbol{k}_\bullet$)



- Non-local CC theorem: **non-local** CC signal ⊂ **bulk-cut** diagram



Pair-production from bkg